Automation script project

Module 6 - Lesson 1

Overview

Automation scripts replace repetitive manual work. A well-structured script takes inputs, performs a task, and reports results clearly.

Key Concepts

- Start from a real pain point: what do you do by hand more than once?
- Accept inputs via arguments (argparse or sys.argv) and configuration files.
- Validate inputs and fail with clear messages early.
- Structure the script with small functions: parse, process, output.
- Log progress so a long run is auditable.
- Handle partial failure: process what works, report what failed.

Example

```
import argparse

parser = argparse.ArgumentParser(description="Rename files by pattern")
parser.add_argument("pattern")
parser.add_argument("--dry-run", action="store_true")
args = parser.parse_args()

# with args.pattern and args.dry_run, walk files and rename
```

Summary

An automation script turns repeated manual steps into one command. Clear input handling, small functions, and logging keep it robust for daily use.

Practice Checklist

- Identify a repetitive task and script it end to end.
- Add a --dry-run flag that prints what would happen.
- Write a test for the core processing function.